Chapter 6: FIR Filter Design

Introduction:

Filters are a particularly important class of linear time-invariant systems. In the design of frequency selective filters, the desired filter characteristics are specified in the frequency domain in terms of the desired magnitude and phase response of the filter. In the filter design process, we determine the coefficients of a causal FIR or IIR filter that closely approximate the desired frequency response specifications.

In practice, FIR filters are employed in filtering problems where there is a requirement for a **linear-phase** characteristic within the passband of the filters. As a general rule, an IIR filter has lower sidelobes in the stopband than a FIR filter having the same number of parameters. For this reason, if some phase distortion is either tolerable or unimportant, an IIR filter is preferable, primarily because its implementation involves fewer parameter, require less memory & has lower computational complexity.

Design of Digital FIR filters:

- If a **linear phase** characteristic is required then FIR design, else FIR or IIR filter design.
- As a general rule, an IIR filter has lower sidelobes in the stopband than a FIR filter having the same number of parameters.
- IIR implementation involves a fewer parameters, require less memory and has lower computational complexity.

Why FIR filters?

In many digital signal processing applications, FIR filters are preferred over their IIR counterparts. The main advantages of the FIR filter designs over their IIR equivalents are the following:

i. They can have an exact linear phase and the design methods are generally linear.
ii. There exist computationally efficient realizations for implementing FIR filters. These include both nonrecursive and recursive realizations.
iii. FIR filters realized nonrecursively are inherently stable and free of limit cycle oscillations when implemented on a finite-word length digital system.
iv. They can be realized efficiently in hardware.
v. The filter start-up transients have finite duration.
vi. The output noise due to multiplication roundoff errors in an FIR filter is usually very low and the sensitivity to variations in the filter coefficients is also low.

The main disadvantage of conventional FIR filter designs is that they require, especially in applications demanding narrow transition bands, considerably more arithmetic operations and hardware components, such as multiplier, adders, and delay elements than do comparable IIR filters.

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Causality and its implications:

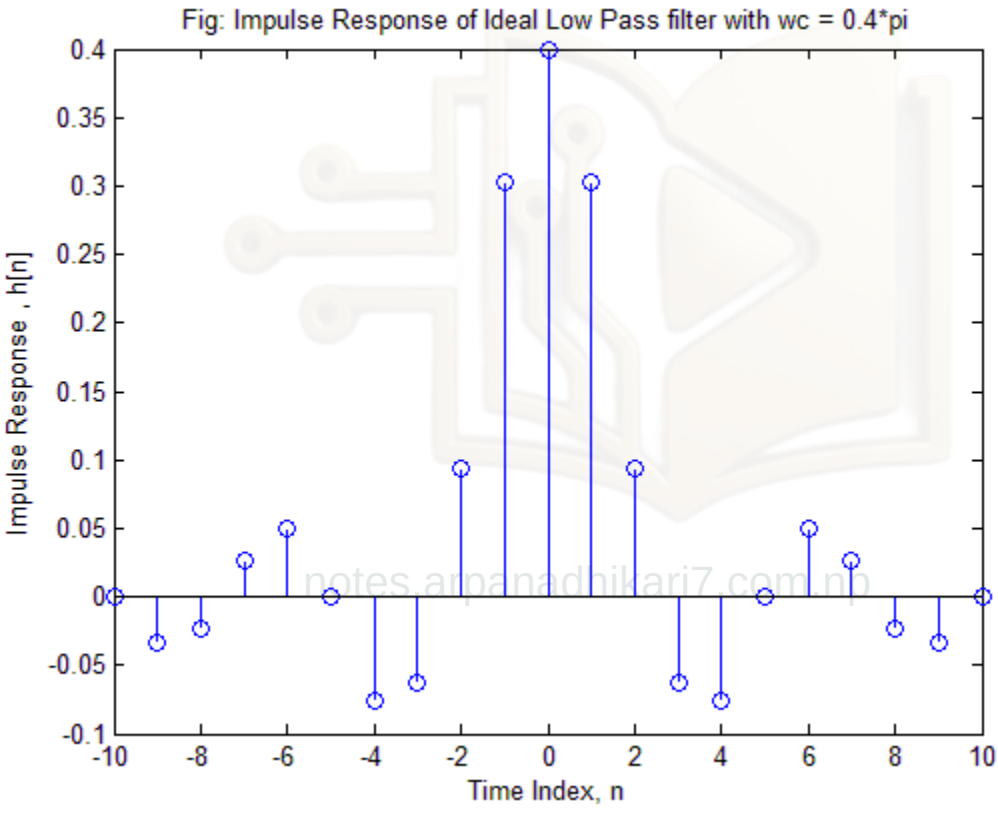
Let us examine the impulse response h[n] of an ideal lowpass filter with frequency response characteristics:

$$H(\omega) = \begin{cases} 1, & |\omega| \le \omega_c \\ 0, & \omega_c < \omega \le \pi \end{cases}$$

The impulse response of this filter is

$$h[n] = \begin{cases} \omega_c/\pi, & n = 0 \\ \frac{\omega_c}{\pi}\frac{sin\omega_c n}{\omega_c n}, & n \neq 0 \end{cases}$$

Which is a Sinc function and it is clear that the ideal lowpass filter is noncausal and hence it cannot be realized in practice.



One possible solution is to introduce a large delay n_0 in h[n] and arbitrarily to set h[n] = 0 for n < n_0. However, the resulting system no longer has an ideal frequency response characteristic. If we set h[n] = 0 for n < n_0, the Fourier series expansion of H(ω) results in the Gibbs Phenomenon.

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Magnitude Response and Phase Response of Digital Filters:

The discrete-time Fourier transform of a finite sequence impulse response h[n] is given by

$$H(e^{j\omega}) = \sum_{n=0}^{M-1} h[n]e^{-j\omega n} = \left|H(e^{j\omega})\right|e^{j\varphi(\omega)}$$

The magnitude and phase responses are given by:

$$M(\omega) = \left|H(e^{j\omega})\right| = \sqrt{(H_R(e^{j\omega})^2 + H_I(e^{j\omega})^2)}$$
$$\varphi(\omega) = \tan^{-1}\left(H_I(e^{j\omega}) \Big/ H_R(e^{j\omega})\right)$$

Where $H_R(e^{j\omega})$ = Re{$H(e^{j\omega})$} & $H_I(e^{j\omega})$ = Im{ $H(e^{j\omega})$}

Filters can have a linear or non-linear phase, depending upon the delay function, namely the phase delay and group delay. The phase and group delays of the filter are given by

$$\tau_p = -\frac{\varphi(\omega)}{\omega} \qquad \& \qquad \tau_g = -\frac{d\varphi(\omega)}{d\omega}$$

Linear phase filters are those filters in which the phase delay and group delay are constants i.e. independent of frequency. Linear phase filters are also called constant time delay filters. For the phase response to be linear

$$\frac{\varphi(\omega)}{\omega} = -\tau, \qquad -\pi \le \omega \le \pi$$
$$\therefore\ \varphi(\omega) = -\omega\tau$$

Where τ is a constant phase delay expressed in number of samples.

$$\varphi(\omega) = \tan^{-1}\frac{H_I(e^{j\omega})}{H_R(e^{j\omega})} = -\omega\tau = \tan^{-1}\frac{-\sum_{n=0}^{M-1} h[n]\sin\omega n}{\sum_{n=0}^{M-1} h[n]\cos\omega n}$$
$$or, \tan(\omega\tau) = \frac{\sum_{n=0}^{M-1} h[n]\sin\omega n}{\sum_{n=0}^{M-1} h[n]\cos\omega n}$$

Simplifying, we get

$$\sum_{n=0}^{M-1} h[n]\sin(\omega\tau - \omega n) = 0$$

And the solution is given by,

$$\tau = \frac{M-1}{2} \quad \& h[n] = h[M-1-n]\ (Symmetric\ Impulse\ response)\ for\ 0 < n < M-1$$

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If these conditions are satisfied, then the FIR filter will have constant phase and group delays and thus the phase of the filter will be linear. The phase and group delays of the linear phase FIR filter are equal and constant over the frequency band. Whenever a constant group delay alone is preferred, the impulse response will be of the form,

$$h[n] = -h[M-1-n] \longrightarrow Antisymmetric\ impulse\ response\ sequence.$$

FIR filter design based on windowed Fourier series:

FIR filters are described by a transfer function that is a polynomial in z^{-1} and require different approaches for their design.

A direct and straightforward method is based on truncating the Fourier series representation of the prescribed frequency response.

The second method is based on the observation that for a length of M FIR digital filter, M distinct equally spaced frequency samples of its frequency response constitute the M-point DFT of its impulse response and hence, the impulse response sequence can be readily computed by applying an inverse DFT to these frequency samples.

Impulse Response of Ideal Filter:

Ideal lowpass filter has zero-phase frequency response

$$H_{LP}(e^{j\omega}) = \begin{cases} 1, & |\omega| \le \omega_c \\ 0, & \omega_c < |\omega| \le \pi \end{cases}$$

The corresponding impulse response,

$$h_{LP}(n) = \begin{cases} \omega_c/\pi, & n = 0 \\ \frac{\omega_c}{\pi}\frac{\sin \omega_c n}{\omega_c n}, & n \neq 0 \end{cases} = \frac{\sin \omega_c n}{\pi n}, \qquad -\infty \le n \le \infty$$

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The impulse response of an ideal lowpass filter is doubly infinite, not absolutely summable, and therefore unrealizable. By setting all impulse response coefficients outside the range $-\tau \le n \le \tau$ equal to zero, we arrive at a finite length non-causal approximation of length $M = 2\tau + 1$, which is shifted to right yield the coefficients of a causal FIR lowpass filter:

$$\hat{h}_{LP}(n) = \begin{cases} \frac{\sin \omega_c(n-\tau)}{\pi(n-\tau)}, & 0 \le n \le M-1 \\ 0, & otherwise \end{cases}$$

This impulse response is desired one and is denoted by $h_d[n]$.

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FIR filter design by windowing:

If $H_d(\omega)$ is desired frequency response and $h_d[n]$ be its corresponding unit sample response. We begin with $H_d(\omega)$ & determine the $h_d[n]$. The relations are given by Fourier transform.

$$H_d(\omega) = \sum_{n=0}^{\infty} h_d[n] e^{-j\omega n} \quad ---(i)$$

Where,

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} \, d\omega \quad ---(ii)$$

Here, the unit sample response from (i) is infinite duration and it is also non-causal and unrealizable. In order to obtain a realizable filter, it must be truncated at some point. Let n = M-1 to yield the FIR filter of length M. i.e.

$$h[n] = \begin{cases} h_d[n], & 0 \le n \le M-1 \\ 0, & otherwise \end{cases}$$

Now, frequency response corresponding to the finite duration sequence is,

$$H(\omega) = \sum_{n=0}^{M-1} h_d[n] e^{-j\omega n} \quad ---(iii)$$

The process of obtaining (iii) from (i) is **windowing**. That is, truncating $h_d[n]$ to length M is equivalent to multiplying $h_d[n]$ by a Rectangular window, defined as

$$w[n] = \begin{cases} 1, & 0 \le n \le M-1 \\ 0, & otherwise \end{cases}$$

So, the unit sample response of FIR filter becomes,

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$$h[n] = h_d[n] w[n] = \begin{cases} h_d[n], & 0 \le n \le M-1 \\ 0, & otherwise \end{cases}$$

note that the multiplication of $h_d[n]$ and $w[n]$ results that the convolution of $H_d(\omega)$ and $W(\omega)$ will be the frequency domain representation of designed FIR filter. i.e.

$$H(\omega) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(v) W(\omega - v) dv$$

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Gibbs Phenomenon:

The causal FIR filters, obtained by simply truncating the impulse response coefficients of the ideal filters given in the previous section exhibit an *oscillating behavior,* in their magnitude response is known as Gibb's Phenomenon.

The oscillation behavior of the magnitude response on both sides of the cut-off frequency is clearly visible in both cases. Moreover, as the length of the filter increased, the number of ripples in both passband and stopband increases with a corresponding decrease in the width of the ripples. However, the heights of the largest ripples, which occurs in both sides of the cut-off frequency remains same independent of the filter length and are approximately 11 percent of the difference between the passband and stopband magnitude of the ideal filters.

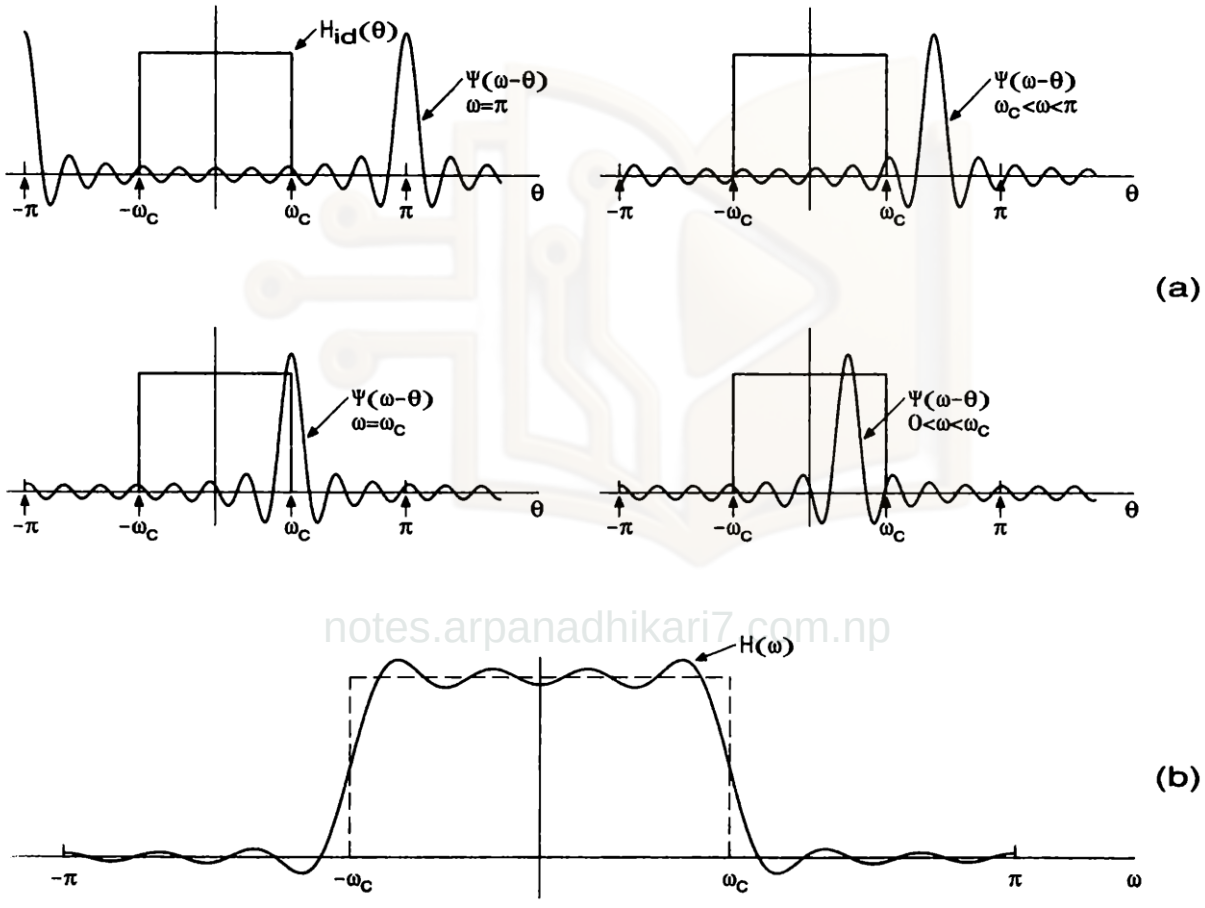


Figure: Illustration of effect of windowing in the frequency domain & Gibbs phenomenon.

The reason behind the Gibb's phenomenon can be explained by considering the truncation operations as multiplication by a finite-length window sequence ω(n) and by examining the windowing process in the frequency domain.

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Rectangular window:

The rectangular window is defined as,

$$w[n] = \begin{cases} 1, & 0 \le n \le M-1 \\ 0, & otherwise \end{cases}$$

Then its Fourier transform is,

$$W(\omega) = \sum_{n=0}^{M-1} w[n]\, e^{-j\omega n}$$

On expansion we get,

$$W(\omega) = e^{-j\omega(M-1)/2} \frac{\sin(\omega M/2)}{\sin(\omega/2)}$$

This window function has a magnitude response

$$|W(\omega)| = \left| \frac{\sin(\omega M/2)}{\sin(\omega/2)} \right|, \qquad -\pi \le \omega \le \pi$$

Here, the width of the main lobe is 4π/M. As the value of M increases the main lobe becomes narrower. However, the sidelobes of |W(ω)| remain unaffected by value of M.

Note: (i) when M is increased, the width of main lobe is decreased and the transition band is reduced. (ii) Attenuation in side lobes is independent of M but it depends upon types of windows. (iii) A window function with minimum stop band, attenuation has maximum main lobe width. So proper window should be chosen in order to achieve a desire stop band attenuation.

Types of Windows:

1. Fixed Window Function:

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Type	**w[n] for 0 ≤ n ≤ M-1**	**Transition width of main lobes**	**Minimum stop-band attenuation**
Rectangular	1	$\frac{4\pi}{M+1}$	-21 dB
Bartlett	$1 - \frac{2\lvert n-\tau\rvert}{M-1}$	$\frac{8\pi}{M}$	-25 dB
Hanning	$0.5\left(1 - \cos\frac{2\pi n}{M-1}\right)$	$\frac{8\pi}{M}$	-44 dB
Hamming	$0.54 - 0.46\cos\left(\frac{2\pi n}{M-1}\right)$	$\frac{8\pi}{M}$	-53 dB
Blackman	$0.42 - 0.5\cos\frac{2\pi n}{M-1} + 0.08\cos\frac{4\pi n}{M-1}$	$\frac{12\pi}{M}$	-74 dB

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Note:

a. The $H_d(\omega)$ of an ideal LP filter is

$$H_d(\omega) = \begin{cases} e^{-j\omega\tau}, & for\ |\omega| \le \omega_c \\ 0, & elsewhere \end{cases}$$

b. The phase delay $\tau = \frac{M-1}{2}$. We should take odd value of M.

c. The order of the filter can also be obtained from transition width of the main lobe.

2. Adaptive Window

In fixed window function, it depends only on the length of the window i.e. M. In order word, the width of the main lobe and attenuation of sidelobes depends only upon the length 'M' of the window. They cannon be controlled independently.

a. Kaiser window:
Kaiser window allows separate control of width of the main lobe and attenuation of sidelobes. The Kaiser window is defined as,

$$w[n] = \frac{I_0\left[\beta\left(\sqrt{1-\left({}^{(n-\tau)}/_{\tau}\right)^2}\right)\right]}{I_0(\beta)}\ for\ 0 \le n \le M-1$$

β = adjustable parameter, $I_0(x)$ = zeroth order Bessel function

$$I_0(x) = 1 + \sum_{r=1}^{\infty}\left[\frac{\left({}^{x}/_{2}\right)^r}{r!}\right]^2$$

β controls the minimum attenuation $\alpha_s = -20\log_{10}\delta_s$, in the stopband of the windowed filter response.

$$\beta = \begin{cases} 0.1102(\alpha_s - 8.7) & \text{for } \alpha_s > 50 \\ 0.5842(\alpha_s - 21)^{0.4} + 0.07886(\alpha_s - 21) & \text{for } 21 < \alpha_s \le 50 \\ 0 & for\ \alpha_s < 21 \end{cases}$$

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The order of filter M is estimated by using the formula

$$M = \begin{cases} \frac{\alpha_s - 7.95}{14.36\Delta f} + 1 & \text{for } \alpha_s > 21 \\ \frac{0.9222}{\Delta f} + 1 & \text{for } \alpha_s \le 21 \end{cases}$$

$$\Delta f = f_s - f_p = \frac{\omega_s - \omega_p}{2\pi}\ (transition\ width)$$

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FIR filters design by Frequency Sampling method:

In this method

- A set of samples is found from the desired frequency response, which is considered as DFT coefficients.
- Then IDFT of these samples are calculated which gives filter coefficients.

$$H_d(\omega) \xrightarrow{sample} H(k) \xrightarrow{IDFT} h(n)$$

The $H_d(\omega)$ is sampled at

$$\omega_k = \frac{2\pi}{M}(k+\alpha), \qquad k = 0,1,2,---,K$$

$$K = \begin{cases} \dfrac{M-1}{2} & if\ M\ odd \\ \dfrac{M}{2} - 1 & if\ M\ even \end{cases}$$

If $\alpha = 0$, Type I design, else if $\alpha = ½$, Type II design.

Type I Design of frequency sampling method:

Consider the design of the FIR filter whose desired frequency response is denoted by $H_d(\omega)$. This frequency response is sampled uniformly at M points. Such frequency samples are given at,

$$\omega_k = \frac{2\pi}{M}k, \;\; k = 0,1,2,3,\dots,M-1$$

Such sampled desired frequency response is a Discrete Fourier Transform, it can be denoted by

$$H(k) = H_d(\omega)|_{\omega=\omega_k}, \;\; k = 0,1,2,\dots,M-1$$

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$$or\ H(k) = H_d\left(\frac{2\pi k}{M}\right), \;\; k = 0,1,2,\dots,M-1$$

Hence H(k) is M-point DFT. By taking inverse DFT of H(k), we get h[n]. This h[n] is unit sampled response of FIR filter.

$$h[n] = \frac{1}{M}\sum_{k=0}^{M-1} H(k)e^{\frac{j2\pi kn}{M}}, \quad n = 0,1,2,3,\dots,M-1$$

hence, the unit sample response of FIR filter of length M is obtained using frequency sampling method. The above impulse response can be simplified for linear phase condition as,

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$$h[n] = \frac{1}{M}\left[H(0) + 2\sum_{k=1}^{K} Re\left\{H(k)e^{\frac{j2\pi kn}{M}}\right\}\right]$$

$$K = \begin{cases} \dfrac{M-1}{2} & if\ M\ odd \\ \dfrac{M}{2} - 1 & if\ M\ even \end{cases}$$

Derive an expression for system function if the unit sample response h[n] is obtained using frequency sampling method.

Solution: ➔

The system function H(z) is given as z-transform of h[n] i.e.

$$H(z) = \sum_{n=0}^{M-1} h[n]z^{-n}$$

Substituting for h[n]

$$H(z) = \sum_{n=0}^{M-1}\left[\frac{1}{M}\sum_{k=0}^{M-1} H(k)e^{\frac{j2\pi kn}{M}}\right] z^{-n}, n = 0, 1, 2, 3, \ldots, M-1$$

Interchanging the order of summation in above equation, we get

$$H(z) = \frac{1}{M}\sum_{k=0}^{M-1}\left[H(k)\sum_{n=0}^{M-1} e^{\frac{j2\pi kn}{M}} z^{-n}\right]$$

The inner summation is in the form of,

$$\sum_{n=N_1}^{N_2} a^n = \frac{a^{N_1} - a^{N_2+1}}{1-a}$$

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$$e^{\frac{j2\pi kn}{M}} z^{-n} = \left(e^{\frac{j2\pi k}{M}} z^{-1}\right)^n$$

Now applying above result to the summation,

$$\sum_{n=0}^{M-1}\left(e^{\frac{j2\pi k}{M}} z^{-1}\right)^n = \frac{1-\left(e^{\frac{j2\pi k}{M}} z^{-1}\right)^M}{1-e^{\frac{j2\pi k}{M}} z^{-1}} = \frac{1-e^{j2\pi k} z^{-M}}{1-e^{\frac{j2\pi k}{M}} z^{-1}} = \frac{1-z^{-M}}{1-e^{\frac{j2\pi k}{M}} z^{-1}}$$

$$\therefore H(z) = \frac{1-z^{-M}}{M}\sum_{k=0}^{M-1}\left[\frac{H(k)}{1-e^{\frac{j2\pi k}{M}} z^{-1}}\right]$$

This equation gives the system function H(z).

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